

BJP TECHNOLOGIES

(T) LIMITED

SESSION 13

WORK REPORT

Date: 10 May 2026

Project: bjptechnologies.co.tz

Phase 6 — Polish & Launch

1. GitHub Repository About Section

Problem:

The repository had no description, website URL, or topic tags — presenting a blank 'About' panel to anyone visiting the repo on GitHub.

Resolution:

The About section was populated using `gh repo edit` via the GitHub CLI:

Field	Value
Description	Corporate website for BJP Technologies (T) Limited — IT solutions company in Tanzania. Built with Django 6, MySQL, and deployed on cPanel.
Website	https://bjptechnologies.co.tz
Topics	django, python, mysql, cpanel, corporate-website, tanzania

2. Homepage Counter Bug Fix

Problem:

The homepage stats section displayed four counters: **7** IT Service Lines, **6** Industries Served, **24/7** Monitoring & Support, and **100%** Dedicated to Your Success. The first two animated correctly. The last two displayed as **0**.

Root cause: the `counter-up.js` plugin reads each `.counter` span's text content and parses it as a number. Values containing non-numeric characters (24/7, 100%) produce NaN, which the plugin renders as **0**, overwriting the correct value.

Fix:

Modified `static/js/main.js` — the `counterUp` function now iterates each `.counter` span and applies the animation only when the value matches a purely numeric pattern (`/^[\\d, ]+(\\. [\\d]+)?$/`). Non-numeric values are displayed as-is without animation.

Branch merged: `fix/homepage-counter-non-numeric`

3. Automated cPanel Deployment

Problem:

Every deployment required manual intervention: log into cPanel, navigate to Git Version Control, and click Pull. There was no automation triggered by merges to the main branch.

Goal: Auto-deploy on every merge to main.

Approaches Investigated and Ruled Out:

Approach	Reason Failed
SSH into cPanel (port 22)	Port blocked by hosting provider

Approach	Reason Failed
cPanel UAPI — port 2083	OpenResty proxy returns HTTP 415 for all external calls
cPanel JSON API	Same proxy, same HTTP 415 error
SSH deploy key for private repo	No terminal access to configure ~/.ssh/config
PAT in Git Version Control URL	cPanel UI explicitly rejects passwords in clone URLs

Final Solution: GitHub Webhook + Django Endpoint

Since the cPanel API is blocked from external IPs but the live Django site is publicly reachable, a webhook-based approach was implemented:

Step	Action
1	Merge PR into main on GitHub
2	GitHub fires a POST webhook to https://bjptechnologies.co.tz/deploy/webhook/
3	Django validates the HMAC-SHA256 signature using GITHUB_WEBHOOK_SECRET
4	Django spawns scripts/deploy.sh as a fully detached background process
5	deploy.sh runs: git pull origin main
6	deploy.sh runs: pip install -r requirements.txt
7	deploy.sh runs: python manage.py migrate --no-input
8	deploy.sh runs: python manage.py collectstatic --no-input
9	deploy.sh touches tmp/restart.txt — Passenger restarts with new code

Files Created / Modified:

File	Action	Notes
apps/core/webhook.py	Created	Django webhook view with HMAC-SHA256 validation
scripts/deploy.sh	Created	Shell script: git pull, pip, migrate, collectstatic, restart
config/urls.py	Modified	Registered /deploy/webhook/ route
.github/workflows/deploy.yml	Modified	Removed broken deploy job — tests only now
.cpanel.yml	Created	cPanel post-deploy config (fallback for manual pulls)
.env.example	Modified	Added GITHUB_WEBHOOK_SECRET variable

Server-Side Setup (Manual, One-Time):

- Generated a cryptographically random GITHUB_WEBHOOK_SECRET token
- Added GITHUB_WEBHOOK_SECRET to the server .env file via cPanel File Manager
- Configured GitHub webhook: payload URL, secret, push events only, SSL verification on
- Performed one manual git pull via cPanel Git Version Control to bootstrap the endpoint

Result:

Webhook delivery confirmed returning HTTP 200. Deploy script confirmed running on the server — log output visible at /tmp/bjp_deploy.log. Automated deployment is fully operational.

4. Current Project State

Area	Status	Notes
Phase 6 — Polish & Launch	In Progress	Ongoing
Automated deployment	Complete	GitHub webhook fully operational
Page load speed	Pending	Target: under 3 seconds
Image optimisation	Pending	All images to be compressed
Final security audit	Pending	No secrets, no DEBUG, no SQL exposure
Go-live sign-off	Pending	Awaiting all Phase 6 items
Private repository	Blocked	Host blocks SSH + PAT auth — needs provider help or VPS

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